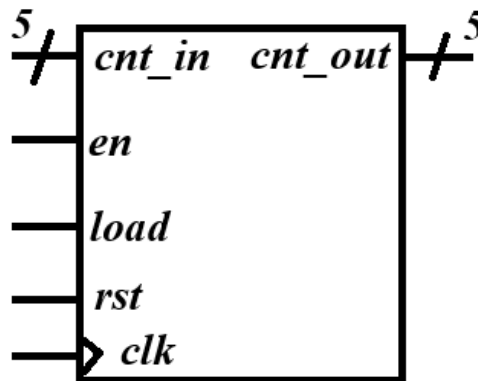


Chapter 1: Basic Digital Design and Testbench

Exercise 1: In this exercise, you create a counter design as the specification and write testbench to verify it. The design is shown in the figure below.



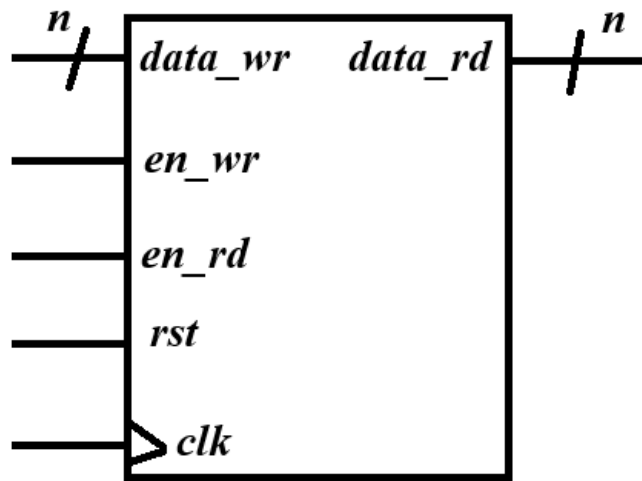
Specifications

- The counter is clocked on the rising edge of *clk*.
- *rst* is active high.
- *cnt_in* and *cnt_out* are both 5-bit signals.
- If *rst* is high, output will become zero.
- If *load* is high, the counter is loaded from the input *cnt_in*.
- Otherwise, if *en* is high, *cnt_out* is incremented and *cnt_out* is unchanged.

Your testbench should show the following results.

- At time 20 *rst*=0 *load*=1 *enab*=1 *cnt_in*=10101 *cnt_out*=10101
- At time 30 *rst*=0 *load*=1 *enab*=1 *cnt_in*=01010 *cnt_out*=01010
- At time 40 *rst*=0 *load*=1 *enab*=1 *cnt_in*=11111 *cnt_out*=11111
- At time 50 *rst*=1 *load*=1 *enab*=1 *cnt_in*=11111 *cnt_out*=00000
- At time 60 *rst*=0 *load*=1 *enab*=1 *cnt_in*=11111 *cnt_out*=11111
- At time 70 *rst*=0 *load*=0 *enab*=1 *cnt_in*=11111 *cnt_out*=00000

Exercise 2: In this exercise, you create a Stack design as the specification and write testbench to verify it. The design is shown in the figure below.



Specifications

- Parameterized for width and depth: Depth restricted to power of 2
- Asynchronous high-active reset
- Write/read synchronous to rising clock
- Data written later will be read out first
- Ignore writing when full
- Ignore reading when empty

After completing the design, present the test scenarios and then write a testbench to verify it.